

Portfolio Performance & Risk

A diversified allocation against a U.S. 60/40 benchmark

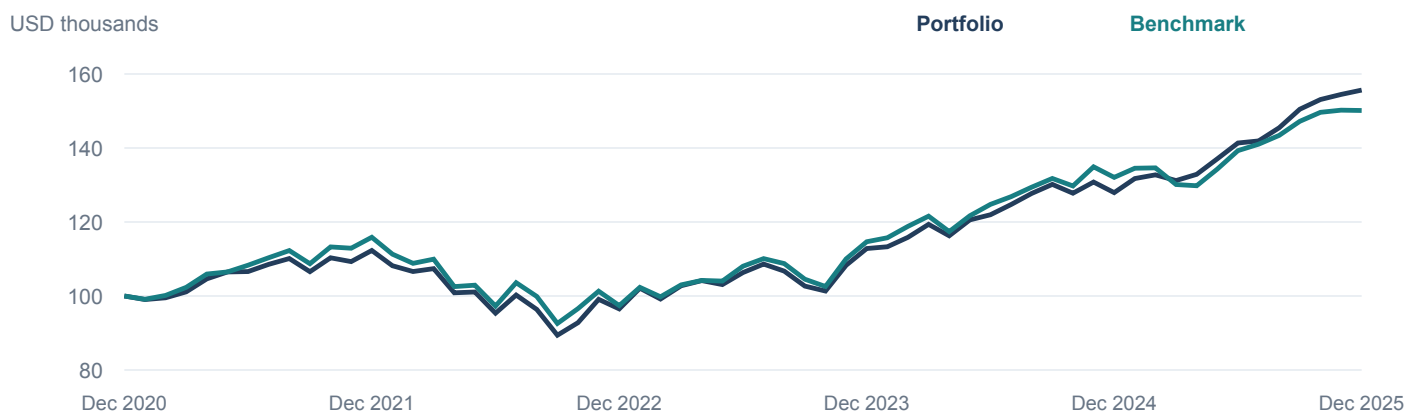
January 2021 - December 2025 / USD / Prepared by Trimanjeev Sawhney

Research finding

The illustrative portfolio compounded at **9.25%** annually, versus 8.47% for the benchmark. A \$100,000 starting allocation ended at **\$155,651**, a \$5,516 advantage. Volatility was modestly lower, but the maximum month-end drawdown was slightly deeper. The result supports evaluating diversification across several risk measures.

Five-year measure	Portfolio	Benchmark
Annualized return (CAGR)	9.25%	8.47%
Cumulative total return	55.65%	50.13%
Annualized volatility	10.49%	10.86%
Maximum month-end drawdown	-20.39%	-20.10%

Growth of \$100,000



Calendar-year performance

Year	Portfolio	Benchmark	Difference (pp)
2021	12.30%	15.87%	-3.57
2022	-14.06%	-15.94%	+1.88
2023	16.91%	17.72%	-0.82
2024	13.39%	15.17%	-1.78
2025	21.66%	13.69%	+7.98

Portfolio targets: 40% IVV, 20% IEFA, 30% AGG and 10% IAU. Benchmark: 60% IVV and 40% AGG. Both rebalance quarterly; holdings drift between resets. The comparison uses provider monthly NAV total returns [1-4], with distributions reinvested. This is a retrospective hypothetical allocation, not an actual account or investment-club track record.

Risk and performance drivers

Diversification reduced volatility, but did not remove downside risk



Both strategies peaked in December 2021 and reached their deepest month-end decline in September 2022. The portfolio lost 20.39% from its peak, versus 20.10% for the benchmark. It recovered the peak in December 2023; the benchmark recovered in February 2024. Monthly observations omit interim lows, so daily drawdowns could be larger.

Return correlations

Monthly returns	IVV	IEFA	AGG	IAU
IVV	1.00	0.80	0.63	0.07
IEFA	0.80	1.00	0.75	0.32
AGG	0.63	0.75	1.00	0.37
IAU	0.07	0.32	0.37	1.00

Pearson correlations across 60 monthly observations. IVV and IEFA moved together strongly (0.80), while IAU had low correlation with IVV (0.07). AGG was positively correlated with both equity funds over this sample. These historical relationships can change.

What drove the results

2022 exposed the limits of stock-bond diversification. IVV returned -18.14%, IEFA -15.13% and AGG -13.05%. The portfolio lost 14.06% for the calendar year, versus 15.94% for the benchmark. The Federal Reserve raised its target rate to 4.25%-4.50% in December 2022 while describing inflation as elevated [5]. This provides market context; the model does not estimate causal macroeconomic effects.

2025 changed the relative outcome. The portfolio returned 21.66%, exceeding the benchmark by 7.98 percentage points. IEFA gained 31.84% and IAU 64.60%; their modeled contributions were 6.07 and 5.76 percentage points of the portfolio's annual return. The World Gold Council described strong investment demand in 2025 [6]. That context does not establish that these gains will persist.

Concentration remains in equities. U.S. and developed international equities carry 60% of the target allocation and approximately 80.8% of portfolio variance contribution. Four funds therefore do not imply equal risk sources. Each risk share is the covariance of a holding's monthly weighted return contribution with portfolio return, divided by portfolio variance; this reflects realized weight drift.

Portfolio return contributions are each holding's cumulative dollar profit or loss during the year divided by portfolio value at the start of that year. They sum to portfolio annual return before rounding. This is holding-level contribution, not a formal attribution of excess return versus the benchmark.

Allocation review and methodology

Illustrative rebalance after 31 December 2025

Fund / exposure	Target	End weight	Buy / (sell), USD
IVV / U.S. equity	40%	39.69%	\$477.87
IEFA / Developed ex-U.S./Canada	20%	20.13%	(\$207.64)
AGG / U.S. bonds	30%	29.29%	\$1,100.42
IAU / Gold	10%	10.88%	(\$1,370.65)

Gold finished at 10.88% against a 10% target, while bonds fell to 29.29% against 30%. Restoring targets would sell \$1,578.28 and buy the same amount, equal to 1.01% one-way turnover. These hypothetical dollar trades are outside the return measurement window and exclude costs, taxes, share rounding and execution prices.

Decision framework

The allocation delivered a higher five-year compound return and slightly lower volatility, with a marginally deeper drawdown. It lagged the benchmark in three of five calendar years. Maintain explicit target weights and a consistent review schedule; assess costs and investor constraints before changing exposures. Strong recent gold and international-equity returns alone are insufficient evidence to raise their strategic weights.

Calculation conventions

Portfolio accounting: start with \$100,000 at 31 December 2020, fully invested and without external flows. Apply each asset's monthly return to its opening dollar position. Reset both strategies to target weights before January, April, July and October returns; otherwise retain prior closing holdings. Rebalancing is assumed costless and fractional holdings are permitted.

Risk and return: compound 60 monthly returns; CAGR equals ending wealth / starting wealth raised to 1/5, less one. Volatility is the sample standard deviation of monthly returns multiplied by the square root of 12. Drawdown compares wealth with its prior running peak, including initial capital. The 0.79 percentage-point CAGR gap is a difference in annualized returns, not alpha.

Scope and reproducibility

The allocation was selected retrospectively for analysis and was not optimized. The benchmark has different geographic, bond and gold exposures, so excess return cannot be interpreted as manager skill. NAV returns reflect fund expenses and reinvestment of distributions; taxes, trading costs and NAV-to-market-price execution differences are excluded. All results are in USD, without CAD conversion. Correlations and risk estimates reflect only this five-year sample.

The supplied R analysis reproduces the reported statistics and monthly holdings. The package includes extracted returns and a source manifest; original provider downloads were retained for validation. Provider percentage-return precision varies from two to six decimal places and is preserved in calculations. Display rounding can produce small differences. This static report should be regenerated after changing inputs.

Sources

Monthly NAV total returns: BlackRock / iShares performance downloads, retrieved 8 September 2026. [1] IVV. [2] IEFA. [3] AGG. [4] IAU.

[5] Federal Reserve, FOMC statement, 14 December 2022.

[6] World Gold Council, Gold Demand Trends: Full Year 2025, 29 January 2026.